

ATAI Aircraft – 60 Required Engineering Documents

ATAI Aircraft – Required Engineering Documentation (60 Documents)

Purpose

This document lists the core engineering and development documents typically required for designing, validating, and building a large-scale aerospace platform such as the ATAI Aircraft system. The list follows common aerospace engineering practices used in major aircraft and aerospace programs.

1. System Concept Document
2. System Architecture Document
3. Master Technical Specifications
4. Aircraft General Arrangement
5. Aircraft Configuration Definition
6. Mission Requirements Document
7. Operational Concept (CONOPS)
8. Requirements Traceability Matrix
9. Engineering Assumptions Document
10. Design Philosophy Document
11. Aerodynamic Analysis Report
12. Lift and Buoyancy Calculations
13. Wind Resistance Analysis
14. Flight Stability Study
15. CFD Simulation Report

16. Flight Envelope Definition
17. Control Authority Analysis
18. Hover Performance Analysis
19. Crosswind Performance Study
20. Flight Safety Analysis
21. Structural Design Report
22. Structural Load Analysis
23. Structural Frame Layout
24. Materials Selection Report
25. Structural Safety Factor Analysis
26. Fatigue Life Assessment
27. Finite Element Analysis (FEA)
28. Structural Failure Modes (FMEA)
29. Payload Structural Integration
30. Landing Load Analysis

31. Helium Hull Design Document
32. Gas Containment System Design
33. Buoyancy Control System
34. Pressure Management System
35. Gas Safety Analysis
36. Propulsion System Architecture
37. Distributed Fan Layout
38. Motor Power Specification
39. Propulsion Control System
40. Energy System Architecture
41. Electrical Power System Design
42. Power Distribution Architecture
43. Battery / Energy Storage System
44. Redundancy and Backup Power
45. Flight Control System Architecture

46. Avionics System Design
47. Navigation System Design
48. Sensor Integration Plan
49. Landing System Engineering
50. Landing Leg Structural Design
51. Ground Stability System
52. Shock Absorption System
53. Manufacturing Architecture
54. Assembly Process Plan
55. Supply Chain Strategy
56. Cost Estimation Model
57. Risk Assessment Report
58. Certification Strategy
59. Testing and Validation Plan
60. Development Roadmap

This document serves as a reference checklist for the engineering documentation required during the conceptual design, detailed engineering, validation, and industrialization phases of the ATAI Aircraft platform.